ROHAN PANT

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SUMMARY

AI Engineer and M.S. Data Science student at Rutgers University with 3+ years of production software experience, building LLM applications in Python and TypeScript. Built a RAG chatbot on a local-only LLM stack for Rutgers Medical School and the LLM-as-judge evaluation framework that selected its model; at Cohesity, shipped an ML defect-triage system that cut debugging turnaround 37%. Hands-on with tool-using agents, RAG, and multi-model orchestration across Claude, Gemini, and open-weight models.

PROFESSIONAL EXPERIENCE

Rutgers University - New Brunswick — **New Brunswick, NJ**
Software Developer — **Jan 2026 - Present**

- Built a production RAG chatbot for Rutgers Medical School's clinical guide app on a local-only LLM stack so sensitive data stays on university infrastructure, serving quantized GGUF models through an OpenAI-compatible API.
- Designed the LLM evaluation framework used to choose the production model: benchmarked Llama 3.1 8B, Qwen 2.5 7B, Phi-3.5 Mini, and Mistral 7B on 14 prompts across 9 categories with rule-based and LLM-as-judge scoring.
- Scored RAG retrieval hit rate, citation validity, and latency per model; Llama 3.1 8B led the benchmark at 86% with the best local RAG tradeoff, while cloud 70B inference ran ~40x faster at the same hit rate.
- Built the production deployment pipeline for two live Next.js and TypeScript apps on GitLab CI/CD and AWS EC2, with Docker image builds, health-check smoke tests, a manual promotion gate, and migrations run before launch.

Peregrine — **Remote**
AI Engineering Intern — **May 2026 - Aug 2026**

- Built Scail, an AI search visibility audit engine in Python: a 28-step pipeline with dependency-ordered execution, per-step persistence to Supabase, exponential-backoff retries, and resume-on-retry that skips completed steps.
- Fanned out 30 queries concurrently across 5 AI platforms (ChatGPT, Claude, Gemini, Perplexity, Google AI Overviews) with ThreadPoolExecutor, with Claude Sonnet generating queries and Gemini Flash parsing answers.
- Developed LeadGenAgent, an idempotent multi-step agent chaining prospect discovery, contact enrichment, email verification, scorecard generation, and Netlify deployment across the Google Maps, Apollo, and MillionVerifier APIs.
- Implemented a weighted 0-100 visibility score, automated PDF and Excel deliverables, per-run cost tracking with budget caps and Slack alerts, and a FastAPI control panel, covered by 176 pytest tests.

Cohesity — **Pune, India**
Member of Technical Staff 2 — **Jul 2022 - Jul 2025**

- Engineered a production Auto-Triage ML system in Python (Random Forest and SVM classifiers) that sorted defects into Product, Automation, and Environment categories, reducing debugging turnaround time by 37%.
- Built backend workflows to process and analyze 150k+ system log events using NLP-based pattern detection, enabling faster root-cause analysis across distributed systems.
- Developed and maintained a full Jenkins CI/CD pipeline; built a Grafana dashboard for product resource monitoring, reducing manual QA effort by 20%.

PROJECTS

Patchwork: Bug-to-Regression-Test Agent (Vercel x Google DeepMind Hackathon, Mar 2026)

- Built a GitHub App that turns bug reports into regression tests: on each new issue it reads the repo tree and existing tests via Octokit, has Gemini 2.5 Pro write a matching test, and opens a PR with a confidence rating.
- Moved generation into background work with Vercel waitUntil to stay inside GitHub's 10-second webhook timeout, and logged runs to Supabase for a live dashboard (Next.js, TypeScript, Vercel AI SDK).

Agentically: Agentic RAG for Healthcare Compliance

- Built an agentic RAG system on the Anthropic Claude API with tool use that routes each query across 3 retrieval paths (semantic vector search, exact citation lookup, aggregation), firing tools in parallel for hybrid questions.
- Routed exact standard IDs (e.g. QM.1) to direct lookup instead of embeddings, since bare IDs carry almost no semantic weight in vector search and fail to retrieve.
- Indexed 1,919 regulatory standard sections in MongoDB Atlas Vector Search with 1024-dimension Voyage AI embeddings, auto-embedded on insert via Atlas Triggers, behind a Next.js chat UI on Vercel answering in 3 to 8 seconds.

EDUCATION

Rutgers University — **New Brunswick, NJ**
Master of Science in Data Science ***(GPA: 4.0/4.0)*** — Sep 2025 - May 2027

University of Petroleum and Energy Studies — **Dehradun, India**
Bachelor of Technology in Computer Science and Engineering (Big Data) — Aug 2018 - May 2022

SKILLS

Agentic AI: Tool use, multi-step agents, multi-model orchestration, LLM-as-judge evaluation, prompt engineering
LLMs & Serving: Claude, Gemini, Llama 3.1, Qwen 2.5, Groq, OpenRouter, llama-cpp-python, GGUF, Vercel AI SDK
RAG & Vector DBs: RAG pipelines, semantic search, Voyage AI and MiniLM embeddings, MongoDB Atlas Vector Search, Qdrant
Languages: Python, TypeScript, JavaScript, SQL
Backend & Frameworks: FastAPI, Node.js, Express, Next.js, React, REST APIs, Supabase (PostgreSQL), pytest
Cloud & DevOps: AWS (EC2, S3), Docker, GitLab CI/CD, Jenkins, GitHub Actions, Vercel, Git